

SN54LS189A, SN54LS219A, SN54LS289A, SN54LS319A SN74LS189A, SN74LS219A, SN74LS289A, SN74LS319A 64-BIT RANDOM-ACCESS MEMORIES

D2417, SEPTEMBER 1980—REVISED FEBRUARY 1985

- Organized as 16 Words of Four Bits Each
- Choice of Buffered 3-State or Open-Collector outputs
- Choice of Noninverted or Inverted Outputs
- Typical Access Time . . . 50 ns

description

These monolithic TTL memories feature Schottky clamping for high performance and a fast chip-select access time to enhance decoding at the system level. A three-state output version and an open-collector-output version are offered for both of the logic choices. A three-state output offers the convenience of an open-collector output with the speed of a totem-pole output; it can be bus-connected to other similar outputs, yet it retains the fast rise time characteristic of the TTL totem-pole output. An open-collector output offers the capability of direct interface with a data line having a passive pull-up.

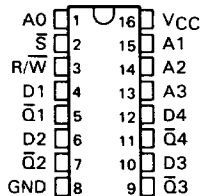
write cycle

Information to be stored in the memory is written into the selected address location when the chip-select ($\overline{S}$) and the write-enable ($R/\overline{W}$) inputs are low. While the write-enable input is low, the memory outputs are off (three-state = Hi-Z, open-collector = high). When a number of outputs are bus-connected, this off state neither loads nor drives the data bus; however, it permits the bus line to be driven by other active outputs or a passive pull-up.

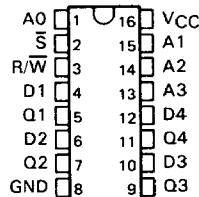
read cycle

Information stored in the memory (see function table for input/output phase relationship) is available at the outputs when the write-enable input is high and the chip-select input is low. When the chip-select input is high, the outputs will be off.

SN54LS189A, SN54LS289A . . . J PACKAGE
SN74LS189A, SN74LS289A . . . J OR N PACKAGE
(TOP VIEW)



SN54LS219A, SN54LS319A . . . J PACKAGE
SN74LS219A, SN74LS319A . . . J OR N PACKAGE
(TOP VIEW)



FUNCTION TABLE

FUNCTION	INPUTS		OUTPUTS			
	CHIP SELECT	WRITE ENABLE	'LS189A	'LS289A	'LS219A	'LS319A
Write	L	L	Z	Off	Z	Off
Read	L	H	Complement of Data Entered	Complement of Data Entered	Data Entered	Data Entered
Inhibit	H	X	Z	Off	Z	Off

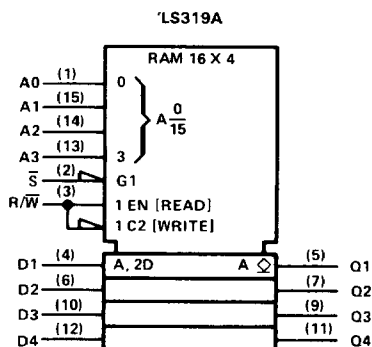
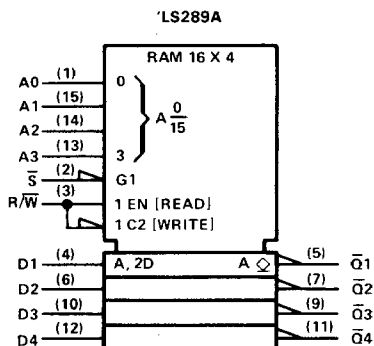
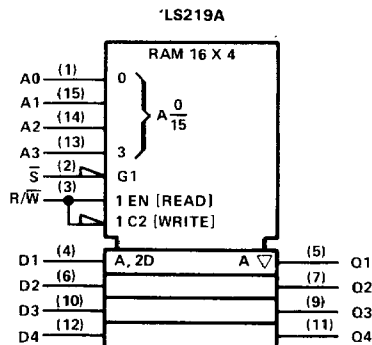
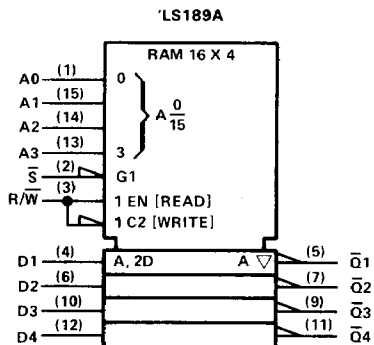
H = high level, L = low level, X = irrelevant, Z = high impedance

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**SN54LS189A, SN54LS219A, SN54LS289A, SN54LS319A
SN74LS189A, SN74LS219A, SN74LS289A, SN74LS319A
64-BIT RANDOM-ACCESS MEMORIES**

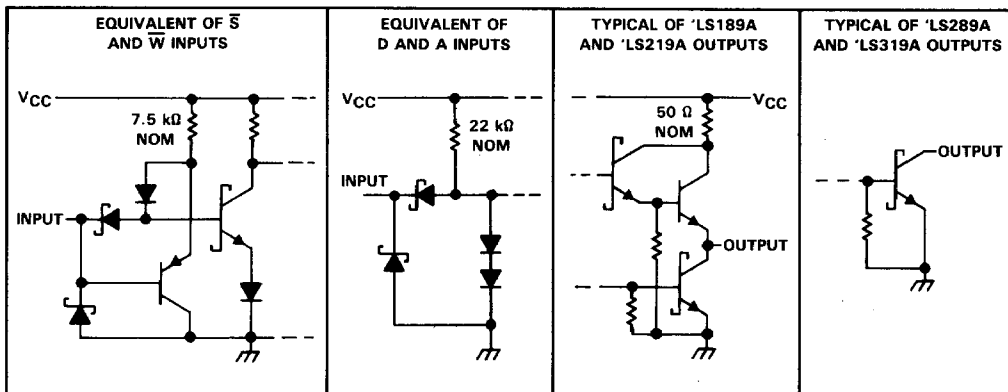
logic symbols



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schematics of inputs and outputs



**SN54LS189A, SN54LS219A, SN54LS289A, SN54LS319A
SN74LS189A, SN74LS219A, SN74LS289A, SN74LS319A
64-BIT RANDOM-ACCESS MEMORIES**

absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

Supply voltage, V_{CC} (see Note 1)	7 V
Input voltage	7 V
Off-state output voltage: 'LS189A, 'LS219A	5.5 V
'LS289A, 'LS319A	7 V
Operating free-air temperature range: SN54LS' Circuits	-55°C to 125°C
SN74LS' Circuits	0°C to 70°C
Storage temperature range	-65°C to 150°C

NOTE 1: Voltage values are with respect to network ground terminal.

recommended operating conditions

		SN54LS189A, SN54LS219A			SN74LS189A, SN74LS219A			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
Supply voltage, V_{CC}		4.5	5	5.5	4.75	5	5.25	V
High-level output current, I_{OH}				-1			-2.6	mA
Low-level output current, I_{OL}				12			24	mA
Width of write pulse (write enable low), $t_{W(wr)}$		100			70			
Setup time	Address before write pulse, $t_{su(ad)}$	0†			0†			ns
	Data before end of write pulse, $t_{su(da)}$	100†			60†			
	Chip-select before end of write pulse, $t_{su(S)}$	100†			60†			
Hold time	Address after write pulse, $t_h(ad)$	0†			0†			ns
	Data after write pulse, $t_h(da)$	0†			0†			
	Chip-select after write pulse, $t_h(S)$	0†			0†			
Operating free-air temperature, T_A		-55 125			0 70			°C

††The arrow indicates the transition of the write-enable input used for reference: † for the low-to-high transition, ‡ for the high-to-low transition.

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SN54LS189A, SN54LS219A, SN74LS189A, SN74LS219A 64-BIT RANDOM-ACCESS MEMORIES WITH 3-STATE OUTPUTS

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS [†]	SN54LS189A SN54LS219A			SN74LS189A SN74LS219A			UNIT
		MIN	TYP [‡]	MAX	MIN	TYP [‡]	MAX	
V _{IH} High-level input voltage		2			2			V
V _{IL} Low-level input voltage				0.7			0.8	V
V _{IK} Input clamp voltage	V _{CC} = MIN, I _I = -18 mA			-1.5			-1.5	V
V _{OH} High-level output voltage	V _{CC} = MIN, V _{IH} = 2 V, V _{IL} = V _{ILmax} , I _{OH} = MAX	2.4	3.1		2.4	3.1		V
V _{OL} Low-level output voltage	V _{CC} = MIN, V _{IH} = 2 V, V _{IL} = V _{ILmax}							
	I _{OL} = 12 mA	0.25	0.4		0.25	0.4		V
	I _{OL} = 24 mA				0.35	0.5		
I _{OZH} Off-state output current high-level voltage applied	V _{CC} = MAX, V _{IH} = 2 V, V _{IL} = V _{ILmax} , V _O = 2.7 V			20			20	μA
I _{OZL} Off-state output current, low-level voltage applied	V _{CC} = MAX, V _{IH} = 2 V, V _{IL} = V _{ILmax} , V _O = 0.4 V			-20			-20	μA
I _I Input current at maximum input voltage	V _{CC} = MAX, V _I = 7 V			100			100	μA
I _{IH} High-level input current	V _{CC} = MAX, V _I = 2.7 V			20			20	μA
I _{IL} Low-level input current	V _{CC} = MAX, V _I = 0.4 V			-0.4			-0.4	mA
I _{OS} Short-circuit output current [§]	V _{CC} = MAX	-30		-130	-30		-130	mA
I _{CC} Supply current	V _{CC} = MAX, See Note 2		35	60		35	60	mA

[†]For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.

[‡]All typical values are at V_{CC} = 5 V, T_A = 25°C.

[§]Not more than one output should be shorted at a time and duration of the short circuit should not exceed one second.

NOTE 2: I_{CC} is measured with the write-enable and chip-select inputs grounded, all other inputs at 4.5 V, and all outputs open.

switching characteristics over recommended operating ranges of T_A and V_{CC} (unless otherwise noted)

PARAMETER	TEST CONDITIONS	SN54LS189A SN54LS219A			SN74LS189A SN74LS219A			UNIT
		MIN	TYP [‡]	MAX	MIN	TYP [‡]	MAX	
t _{a(ad)} Access time from address	C _L = 45 pF, See Note 3	50	90		50	80		ns
t _{a(S)} Access time from chip select (enable time)		35	70		35	60		ns
t _{SR} Sense recovery time		55	100		55	90		ns
t _{pxZ} Disable time from high or low level	from S	30	60		30	50		ns
	from R/W	40	70		40	60		

[‡]All typical values are at V_{CC} = 5 V, T_A = 25°C.

NOTE 3: Load circuits and voltage waveforms are shown in Section 1.

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SN54LS289A, SN54LS319A, SN74LS289A, SN74LS319A **64-BIT RANDOM-ACCESS MEMORIES** **WITH OPEN-COLLECTOR OUTPUTS**

recommended operating conditions

		SN54LS289A, SN54LS319A			SN74LS289A, SN74LS319A			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
Supply voltage, V_{CC}		4.5	5	5.5	4.75	5	5.25	V
High-level output voltage, V_{OH}				5.5			5.5	V
Low-level output current, I_{OL}				12			24	mA
Width of write pulse (write enable low), $t_W(wr)$		100			70			
Setup time	Address before write pulse, $t_{su}(ad)$	01			01			ns
	Data before end of write pulse, $t_{su}(da)$	1001			601			
	Chip-select before end of write pulse, $t_{su}(S)$	1001			601			
Hold time	Address after write pulse, $t_h(ad)$	01			01			ns
	Data after write pulse, $t_h(da)$	01			01			
	Chip-select after write pulse, $t_h(S)$	01			01			
Operating free-air temperature, T_A		-55		125	0		70	°C

†The arrow indicates the transition of the write-enable input used for reference: 1 for the low-to-high transition, 1 for the high-to-low transition.

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS†	SN54LS289A SN54LS319A			SN74LS289A SN74LS319A			UNIT
		MIN	TYP‡	MAX	MIN	TYP‡	MAX	
V_{IH} High-level input voltage		2			2			V
V_{IL} Low-level input voltage				0.7			0.8	V
V_{IK} Input clamp voltage	$V_{CC} = \text{MIN}, I_I = -18 \text{ mA}$			-1.5			-1.5	V
I_{OH} High-level output current	$V_{CC} = \text{MIN}, V_{IH} = 2 \text{ V}, V_O = 2.4 \text{ V}$ $V_{IL} = V_{ILmax}, V_O = 5.5 \text{ V}$			20			20	µA
V_{OL} Low-level output voltage	$V_{CC} = \text{MIN}, V_{IH} = 2 \text{ V}, V_{IL} = V_{ILmax}$ $I_{OL} = 12 \text{ mA}$ $I_{OL} = 24 \text{ mA}$		0.25	0.4		0.25	0.4	V
I_I Input current at maximum input voltage	$V_{CC} = \text{MAX}, V_I = 7 \text{ V}$			100			100	µA
I_{IH} High-level input current	$V_{CC} = \text{MAX}, V_I = 2.7 \text{ V}$			20			20	µA
I_{IL} Low-level input current	$V_{CC} = \text{MAX}, V_I = 0.4 \text{ V}$			-0.4			-0.4	mA
I_{CC} Supply current	$V_{CC} = \text{MAX}$, See Note 2	35	60		35	60		mA

†For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.

‡All typical values are at $V_{CC} = 5 \text{ V}, T_A = 25^\circ\text{C}$.

§Not more than one output should be shorted at a time and duration of the short circuit should not exceed one second.

NOTE 2: I_{CC} is measured with the write-enable and chip-select inputs grounded, all other inputs at 4.5 V, and all outputs open.

switching characteristics over recommended operating ranges of T_A and V_{CC} (unless otherwise noted)

PARAMETER	TEST CONDITIONS	SN54LS289A SN54LS319A			SN74LS289A SN74LS319A			UNIT
		MIN	TYP‡	MAX	MIN	TYP‡	MAX	
$t_a(ad)$ Access time from address	$C_L = 45 \text{ pF}, R_L = 667\Omega$, See Note 3	50	90		50	80		ns
$t_a(S)$ Access time from chip select (enable time)		35	70		35	60		ns
t_{SR} Sense recovery time		55	100		55	90		ns
t_{PLH} Propagation delay time, low-to-high-level output (disable time)		30	60		30	50		ns
		40	70		40	60		

‡All typical values are at $V_{CC} = 5 \text{ V}, T_A = 25^\circ\text{C}$.

NOTE 3: Load circuits and voltage waveforms are shown in Section 1.

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